

Lecture - 15 Resampling Methods

- if we compute a statistic from 1 obs. dataset, but how do we understand how much that statistic would change if we had obtained a diff. random sample.

say, $X_1, \dots, X_n \sim F$, θ
random $\leftarrow \hat{\theta} = T(X_1, \dots, X_n)$

To quantify uncertainty, we want the sampling distribution of $\hat{\theta}$.

- sometimes, there is no problem because probab. theory gives us distribution

$$X_1, \dots, X_n \sim N(\mu, \sigma^2), \quad \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

$$\rightarrow \text{CI will be } \bar{X} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

- we have 1 sample $x_1, \dots, x_n$. The true F is unknown, then obs. data gives approx. distribution c/d empirical distribution.

$$\text{CDF} \Rightarrow F_n(x) = \frac{1}{n} \sum_{i=1}^n I(x_i \leq x) \quad \rightarrow \text{indicator}$$

under F_n , every obs. has probab. $1/n$. $F \rightarrow F_n$
(core idea behind bootstrap).

use the sample as a miniature population.
means, F_n is the best direct nonparametric approx. available from the obs. sample.

- $\theta = t(F)$, but F is unknown. $\hat{\theta} = t(F_n)$
this is c/d plug-in principle.

$$\text{so } F \rightarrow F_n \Rightarrow t(F) \rightarrow t(F_n)$$

In bootstrap we not only estimate itself but distribution of the estimate.

$$L_F(\hat{\theta}) \approx L_{F_n}(\hat{\theta}^*)$$

original
estimator

estimator
calculated from
the bootstrap
sample.

- original obs. sample $\Rightarrow x_1, x_2, \dots, x_n$
bootstrap sample is generated by sampling $\Rightarrow x_1^*, \dots, x_n^*$ from F_n .

For a bootstrap sample b , $\hat{\theta}^*(b) = T(x_1^*(b), \dots, x_n^*(b))$
repeat this B times $\rightarrow \hat{\theta}^{*(1)}, \hat{\theta}^{*(2)}, \dots, \hat{\theta}^{*(B)}$

bootstrap distribution

eg. $x = (2, 4, 5, 7, 9)$ $n = 5$ $\hat{\theta} = \bar{x} = 5.4$
possible bootstrap sample $\Rightarrow (4, 4, 2, 9, 7), \bar{x}^* = 5.2$
" " " $\Rightarrow (2, 9, 9, 5, 5), \bar{x}^* = 6$
(5.2, 6, ...)

- why are we sampling with replacement?

$$P_{F_n}(x^* = x_i) = \frac{1}{n}, \text{ each draw is independent}$$

Some value may not appear, some once, some many.
if we sampled w/o replacement, then every bootstrap will be a permutation of the original data. There will be no meaningful resampling variability. So, bootstrap must sample with replacement.

- There are n draws, each draw $P(x^* = x_i) = \frac{1}{n}$.

No. of appearances N_i satisfies $N_i \sim \text{Bin}(n, \frac{1}{n})$

so any indi. obs. has a 36.8% chance of being missed out. So bootstrap has 63.2% distinct values.

- once we have B replicates, $\hat{\theta}^{*(1)}, \hat{\theta}^{*(2)}, \dots$

avg. bootstrap estimate $\bar{\theta}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}^*(b)$

$$SE_{boot} = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}^*(b) - \bar{\theta}^*)^2}$$

$$SE(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$$

spread of bootstrap estimates $\approx$ sampling uncertainty of $\hat{\theta}$.

$$\boxed{\text{Bias } \hat{\theta} = E_F(\hat{\theta}) - \theta}$$

$$F \rightarrow F_n, E_F(\hat{\theta}) \rightarrow E_{F_n}(\hat{\theta}^*) \approx \bar{\theta}^*$$

$$\boxed{\text{Bias}_{\text{boot}} \hat{\theta} = \bar{\theta}^* - \hat{\theta}}$$

if estimated bias, $(\bar{\theta}^* - \hat{\theta})$
 then corrected estimate $\hat{\theta}_{\text{adj}} = \hat{\theta} - \text{Bias}_{\text{boot}} \hat{\theta}$

$$\boxed{\hat{\theta}_{\text{adj}} = 2\hat{\theta} - \bar{\theta}^*}$$

estimating bias does not mean you have to correct it. As bias correction, will reduce bias, but increase variance.

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

- The bootstrap applies same estimation procedure to resampled versions of the data. If your original estimator is bad. Bootstrap doesn't fix it. Bootstrap measures the variability of an estimator.

- Jackknife $\rightarrow$ leave 1 obs. out

$$\hat{\theta} = T(x_1, \dots, x_n)$$

$i = 1 \dots n$ delete 1 obs. at a time

$$\hat{\theta}_{(-i)} = T(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$

$$\hat{\theta}_{(-1)}, \hat{\theta}_{(-2)} \dots$$

$$\boxed{\hat{\theta}_{(-)} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(-i)}}$$

eg. $x = (2, 4, 5, 7, 9)$ $\hat{\theta} = 5.4$

$$\hat{\theta}_{(-1)} = (4, 5, 7, 9) / 4 = 6.25$$

$$\hat{\theta}_{(-2)} = (2, 5, 7, 9) / 4 = 5.75$$

$$\hat{\theta}_{(-)} = \frac{6.25 + 5.75 + \dots}{5} = 5.4$$

$$SE_{Jack} = \sqrt{\frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(-i)} - \bar{\theta}_{(-)})^2}$$

if deleting ind. observaⁿ barely Δ s the statistic, it is stable.

Eg. $\bar{\theta}_{(-)} = 5.4 \Rightarrow$ find the deviaⁿ $\Rightarrow$ sq. them. $\Rightarrow$ sum

$$SE_{Jack} = \sqrt{\frac{4}{5} (1.825)} \approx 1.208$$

$$\hat{Bias}_{Jack} = (n-1) (\bar{\theta}_{(-)} - \hat{\theta})$$

Eg. $\hat{Bias}_{Jack} = 4 (5.4 - 5.4) = 0$

- Jackknife
delete 1 obs.
n replica
deterministic nature
local perturbation
 $\Downarrow$

makes small, systematic perturbation around the obs. sample $\Rightarrow$ local sensitivity

Bootstrap
sample with replacement
B, B > n
MC
repeated sampling
 $\Downarrow$

creates many pseudo-datasets around empirical distribution
 $\Downarrow$
approx. sampling distribution

- bootstrap works well when statistic is smooth as a funcⁿ of underlying distribution.

Eg. $M = \max(x_1, \dots, x_n)$

$$M^* \leq M$$

- Percentile bootstrap CI

$$\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)}$$

$$q_{\alpha/2}^* \text{ \& \> } q_{1-\alpha/2}^*$$

$$\Rightarrow [q_{\alpha/2}^*, q_{1-\alpha/2}^*]$$

Eg. B = 10K, $q_{0.025}^* = 8.2$, $q_{0.975}^* = 13.7$

Then 95% percentile bootstrap $\Rightarrow [8.2, 13.7]$

→ reverse percentile interval

- basic bootstrap CI $\Rightarrow$ uses error distribution

$$\hat{\theta} - \theta \Rightarrow \hat{\theta}^* - \hat{\theta}$$

$$q_{\alpha/2} \leq \hat{\theta}^* - \hat{\theta} \leq q_{1-\alpha/2}$$

$$\theta \leq \hat{\theta} - q_{\alpha/2} \quad \theta \geq \hat{\theta} - q_{1-\alpha/2}$$

$$\Rightarrow [\hat{\theta} - q_{1-\alpha/2}, \hat{\theta} - q_{\alpha/2}]$$

$$\text{or} \Rightarrow [2\hat{\theta} - q_{1-\alpha/2}^*, 2\hat{\theta} - q_{\alpha/2}^*]$$

Eg. $\hat{\theta} = 10$, $q_{0.025}^* = 8$, $q_{0.975}^* = 13$
Percentile $\Rightarrow [8, 13]$
Basic $\rightarrow$

$$[20 - 13, 20 - 8] \Rightarrow [7, 12]$$

- bootstrap t-interval

$$T = \frac{\hat{\theta} - \theta}{SE(\hat{\theta})}$$

$\Rightarrow$ for each bootstrap

$$T^* = \frac{\hat{\theta}^* - \hat{\theta}}{SE^*}$$

$$t_{\alpha/2}^*, t_{1-\alpha/2}^*$$

$$CI \Rightarrow [\hat{\theta} - t_{1-\alpha/2}^* SE, \hat{\theta} - t_{\alpha/2}^* SE]$$

- studentize $\Rightarrow$ divide an estimator's error by its own estimated SE

$$\frac{\hat{\theta} - \theta}{SE}$$

estimator's error

$\rightarrow$ standardized estimator's error

- frequentist $\Rightarrow$ sampling distribution of $\hat{\theta}$, repeated sampling from F

θ is fixed.

$$X^{(1)}, X^{(2)}, \dots \sim F$$

$$L(\hat{\theta})$$

- bayesian $\rightarrow$ posterior $\pi(\theta|x)$, uncertainty abt. θ after seeing data
 $\pi(\theta|x) \Rightarrow$ posterior distribuⁿ
- bootstrap $\rightarrow$ approximate sampling distribuⁿ of $\hat{\theta}$, resampling from F_n
 $F \approx F_n \quad L_F(\hat{\theta}) \approx L_{F_n}(\hat{\theta}^*)$
- bootstrap confidence intervals is not the same as bayesian credible intervals.